

ITM426, Long Quiz, 2026 Fall

- ITM 426 Engineering Mathematics 26F
- Duration: 90 minutes
- Weights: 10%
- 4 Questions

- Name: _____
- Student ID: _____
- E-mail: _____@seoultech.ac.kr

- Write legibly.
- Justification is necessary unless stated otherwise.
- Partial points are given only sparingly for the most problems because you are expected to 1) carry out proper sanity check and 2) correct your mistake by doing so.

#1. Define what it means for the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ to be *linearly independent*. [5pts]

#2. Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$$

is a basis of a three-dimensional vector space. Prove that

$$\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2 + \mathbf{v}_3, \mathbf{v}_3\}$$

is also a basis of the three-dimensional vector space. [5pts]

#3. Consider the following system of linear equations:

$$x + 2y = 8,$$

$$3x - y = 3.$$

- (a) Express the system in the matrix form $A\mathbf{x} = \mathbf{b}$. [1.5pts]
- (b) Calculate $|A|$. Based on the determinant, determine whether the system has a unique solution. [1.5pts]
- (c) Find the inverse matrix A^{-1} [1pts].
- (d) Using $\mathbf{x} = A^{-1}\mathbf{b}$, find the values of x and y [1pts].

#4. Consider the following system of linear equations, where p and q are real numbers:

$$x + 2y = 3,$$

$$px + 4y = q.$$

- (a) Express the system in the matrix form $A\mathbf{x} = \mathbf{b}$. [1pts]
- (b) Find the determinant $|A|$ in terms of p . Determine the values of p for which A is singular and non-singular. [1pts]
- (c) If A is non-singular, find A^{-1} and obtain the unique solution (x, y) in terms of p and q . [1pts]
- (d) Suppose that A is singular. Determine the condition on q under which the system has infinitely many solutions. [1pts]
- (e) Suppose that A is singular. Determine the condition on q under which the system has no solution. [1pts]